

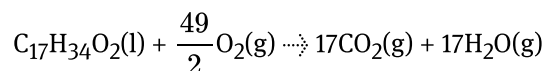
Enthalpy and heat transfers

Learning outcomes

By the end of this section you should be able to:

- Deduce whether a reaction is endothermic or exothermic from a thermochemical equation.
- Analyse data from calorimetry experiments to determine the change in temperature from a chemical reaction.
- Use changes in temperature to calculate the thermal energy, Q , and the enthalpy change of a reaction, ΔH .

Biodiesel is a fuel that is produced from waste vegetable oil or waste animal fats. This is considered a sustainable fuel source because the waste oil takes the place of fossil fuel. Methyl palmitate is a substance derived from vegetable oils that can undergo an exothermic combustion reaction. The release of energy from this reaction can be used for transportation fuels, heating or electricity generation. The combustion reaction of methyl palmitate is:



How might we quantify the energy transferred for specific chemical reactions? What are some ways in which we could measure the energy released in a chemical reaction such as the combustion of methyl palmitate?

Creativity, activity, service

Strand: Service

Learning outcome: Demonstrate engagement with issues of global significance

The combustion of fossil fuels is thought to contribute to global warming and climate change that could have serious consequences for the environment. One possible solution is to increase the use of renewable energy sources  (<https://www.un.org/en/climatechange/what-is-renewable-energy>), such as biofuels and solar cells. As an IB student, can you think of any ways to raise awareness of such issues in your school or local community? One possible CAS experience could be to research the use of renewable energy sources and suggest ways in which their usage could be increased.

Enthalpy and standard enthalpy change of reaction

The amount of heat that a system has is known as its enthalpy (H). The enthalpy of a system is not easily measured but the changes in enthalpy (ΔH) is something we can measure as heat is transferred between system and surroundings. These changes in enthalpy are positive or negative depending whether a reaction absorbs or releases heat.

Exothermic reactions, involve heat being released, indicating a negative change in enthalpy. Endothermic reactions involve heat being absorbed, resulting in a positive change in enthalpy.

Standard enthalpy change of reaction ($\Delta H^\ominus$) refers to the heat transferred at constant pressure and under standard conditions and states. Enthalpy change (ΔH) or standard enthalpy change ($\Delta H^\ominus$) can be included with a chemical reaction providing information about heat transfer for that process. When enthalpy is included beside a chemical reaction, it is known as a thermochemical equation. Next are some examples of thermochemical equations for the combustion of methane and the decomposition of calcium carbonate.

Figure 1 shows an energy profile diagram for these reactions.

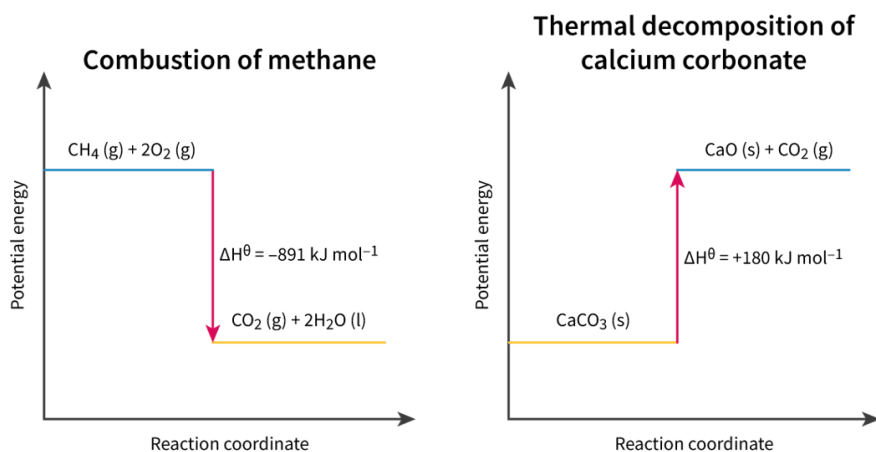
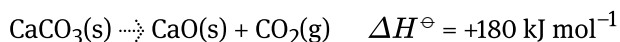
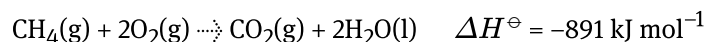


Figure 1. Energy profile diagrams for the combustion of methane and the thermal decomposition of calcium carbonate.

📄 More information for figure 1

Notice how the units of enthalpy are in units of kJ mol^{-1} . This indicates the energy change associated with one mole of a chemical species identified in the chemical reaction. For example, 891 kJ of energy are released for every 1 mole of methane that has undergone combustion and 180 kJ of energy are absorbed for every 1 mole of calcium carbonate that has undergone decomposition.

Thermal energy or heat

Section R1.1.2–3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpy-and-heat-transfers-id-43576/print/>) shows how heat and temperature, while distinct, are related to one another. Energy changes cannot be measured directly in J or kJ, but instead are monitored by temperature changes and then related to thermal energy via the following mathematical relationship shown in **Table 1**.

Table 1. Equations for thermal energy.

Equation	Symbol	Units
$Q = mc\Delta T$	Q = thermal energy (heat absorbed or released)	joules, J
	M = mass (of the species you are measuring the change in temperature of)	grams, g
	c = specific heat capacity (specific to the substance undergoing a change in temperature)	joules per gram per unit Kelvin, $\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$
	ΔT = change in temperature	Celsius or Kelvin $^\circ\text{C}$ or K

Study skills

You may wonder why the units of change in temperature are either in $^\circ\text{C}$ or K. This is because the conversion between the two units involves a difference of 273.15. For example, if a sample has an initial temperature of 0°C that is 273.15 K and ends with a final temperature of 25°C that is 298.15 K, the difference in temperature would be 25°C or 25 K as shown below. This **only** applies to a difference in temperature and not the temperature measurement itself.

ΔT in Celsius	ΔT in Kelvin
$\Delta T = 25^\circ\text{C} - 0^\circ\text{C}$	$\Delta T = 298.15\text{ K} - 273.15\text{ K}$
$\Delta T = 25^\circ\text{C}$	$\Delta T = 25.00\text{ K}$

The specific heat capacity of a substance is the quantity of energy needed to raise 1 g of a substance by 1 K (or 1°C). It is unique to the identity of the substance.

Table 2 shows the specific heat capacity for some common substances.

Table 2. Specific heat capacities for some common substances.

Substance	Specific heat capacity ($\text{J g}^{-1} \text{K}^{-1}$)
Water	4.18
Copper	0.39
Graphite	0.71
Steel	0.47

You can see in **Table 2** that water has a very high specific heat capacity in comparison to the other materials listed. This means it takes a lot of energy to increase the temperature of water and will take a long time to cool down. Conversely, copper does not take as much energy to increase the temperature and will cool down quickly.

Worked example 1

A 150.0 g block of pure copper is heated so that the temperature increases from 18.0°C to 75.0°C. Calculate the heat energy change in kJ if the specific heat capacity of copper is $3.86 \times 10^{-1} \text{ J g}^{-1} \text{ K}^{-1}$.

To calculate the heat energy change we can use the equation:

$$Q = mc\Delta T$$

Substituting the values for the variables given in the question we have:

$$Q = (150.0 \text{ g})(3.86 \times 10^{-1} \text{ J g}^{-1} \text{ K}^{-1})(75.0^\circ\text{C} - 18.0^\circ\text{C})$$

$$Q = 3300.3 \text{ J}$$

The final step would be to convert to kJ by dividing by 1000 and reporting to the appropriate number of significant figures.

$$Q = \frac{3300.3}{1000} \text{ kJ}$$

Therefore, the heat energy change would be.

$$Q = 3.30 \text{ kJ}$$

Calorimetry to determine the enthalpy change of a reaction

Now we know how to relate temperature changes to energy change from a chemical reaction, how can we physically measure these changes in a laboratory setting? Calorimetry is a technique used to measure the heat transfer during a physical or chemical process. A calorimeter is the apparatus used to measure temperature changes to calculate the enthalpy for a reaction. Something as simple as a Styrofoam coffee cup can be sufficient to act as a calorimeter in certain reactions. **Interactive 1** below shows some examples of different calorimeters along with details about the situation you are likely to use each for.

Interactive 1. Types of calorimeter and when to use them.

🔗 More information for interactive 1

Practical skills

Tools 1: Experimental techniques — Measuring variables

Inquiry 1: Exploring and designing — Exploring

You might have noticed in **Interactive 1** that all the calorimeters use water. Why do you think that is? Water has a relatively high heat capacity relative to other substances. This means large amounts of heat produced will result in relatively small temperature changes in water compared to other substances that have low heat capacities. This makes water a practical and safe substance to use for measurement as it will not change state easily or reach temperatures beyond the range of a thermometer.

Additionally, water has a density of 1.00 g cm^{-3} over a large range of temperatures (ranging from 1.00 g dm^{-3} at 0°C to 0.96 g cm^{-3} at 100°C). This allows us to easily relate the volume of the water in the calorimeter to the mass of water changing in temperature.

We can use a calorimeter to measure temperature changes from chemical reactions and calculate the thermal energy transferred in the reaction. To make these quantities specific to a chemical reaction, we can use the thermal energy transferred (Q) to calculate the change in enthalpy for a reaction (ΔH).

You may have noticed that the units of ΔH are kJ mol^{-1} ; this means we need to involve the number of moles involved in a chemical reaction to calculate enthalpy. Additionally, calorimetry involves measuring the temperature of the surroundings, this means that the thermal energy (Q) and the change in enthalpy (ΔH) will be opposite in sign to one another. To account for this, we can calculate the change in enthalpy ΔH using the equation in **Table 3**.

Table 3. Equation for change in enthalpy.

Equation	Symbols	Units
$\Delta H = -\frac{Q}{n}$	ΔH = change in enthalpy	kilojoules per r kJ mol^{-1}
	Q = thermal energy (heat absorbed or released)	joules, J
	n = number of moles of an identified reactant or product in the chemical reaction	moles

Try the **Worked example 2** to calculate the enthalpy of combustion for ethanol.

Worked example 2

A copper calorimeter containing 100.0 g of water is used to measure the enthalpy of combustion of ethanol. Water in the copper calorimeter has an initial temperature of 21.0°C and a final temperature of 67.0°C . The burner containing ethanol had an initial mass of 25.83 g and a final mass of 21.75 g. Calculate the enthalpy of combustion of ethanol in kJ mol^{-1} .

To calculate the enthalpy of combustion of ethanol we need to first calculate the thermal energy that was transferred from the reaction to the water in the calorimeter. To do this, we use:

$$Q = mc\Delta T$$

As water is the entity changing in temperature, we must use the mass and specific heat capacity of water in this equation:

$$Q = (100.0)(4.18)(67 - 21)$$

$$Q = 19\,228\text{ J}$$

To calculate the enthalpy change for combustion of ethanol we need to use the equation:

$$\Delta H = -\frac{Q}{n}$$

Since ethanol is the identified species of interest here, we will need to calculate the number of moles of ethanol. First, we need to determine the mass of ethanol that reacted. We can do this by taking the difference of the final and initial mass of the ethanol burner.

$$m = 25.83 - 21.75$$

$$m = 4.08 \text{ g}$$

The chemical formula of ethanol is $\text{CH}_3\text{CH}_2\text{OH}$, so the molar mass is 46.08 g mol^{-1} .

$$n = \frac{m}{M}$$

$$n = \frac{4.08}{46.08}$$

$$n = 0.0885 \text{ mol}$$

Now we can calculate the change in enthalpy for the combustion of ethanol:

$$\Delta H = -\frac{Q}{n}$$

$$\Delta H = -\frac{19\,228 \text{ J}}{0.0885 \text{ mol}}$$

$$\Delta H = -217\,163 \text{ J mol}^{-1}$$

To convert to kJ

$$\Delta H = -\frac{217\,163}{1000} \text{ kJ mol}^{-1}$$

$$\Delta H = -217 \text{ kJ mol}^{-1}$$

Therefore, the change in enthalpy for the combustion of ethanol is -217 kJ mol^{-1} .

Study skills

When using the equation $Q = mc\Delta T$, a common error made is to use the incorrect mass for the variable m . You should always ask yourself, what is the substance that is changing in temperature? If it is water, then the mass and specific heat capacity should be of water. The variables m , c and ΔT should always be for the same substance.

Determining the enthalpy change involving aqueous reactants

Calculations can sometimes become tricky when determining the change in enthalpy for reactions involving aqueous solutions. Let's consider a chemical reaction involving aqueous copper sulfate and solid zinc as an example.



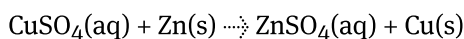
The reaction is exothermic and the release of energy can be monitored by recording the temperature increase over time. But when calculating the thermal energy released, what mass and specific heat capacity would we use in the equation $Q = mc\Delta T$? Because the temperature change measured would be of the aqueous solution, the mass and specific heat capacity would also need to be that of an aqueous solution.

As the reactants are being consumed and the products are forming, their masses are changing throughout the reactions and therefore cannot be used in the $Q = mc\Delta T$ equation. The energy changes instead are monitored by the water solvent.

As water has a density of 1.00 g cm^{-3} , we can easily convert between volumes of water/aqueous solutions and masses when solving for thermal energy. Let's try applying some of these assumptions below to calculate the change in enthalpy for the reaction between zinc and copper(II) sulfate in the **Worked example 3**.

Worked example 3

Calculate the enthalpy change of reaction in kJ mol^{-1} for the reaction of



Given that excess solid zinc reacted with 150 cm^3 of 1.0 mol dm^{-3} copper(II) sulfate resulting in a temperature rise of 43°C .

To calculate the enthalpy of reaction, we need to make use of the following two equations:

$$\Delta H = -\frac{Q}{n} \text{ and } Q = mc\Delta T$$

The aqueous solution would experience the measured temperature rise of 43°C, therefore the mass and specific heat capacity of the aqueous solution can be assumed to be that of water.

As the volume of the aqueous solution is 150 cm³ and water has a density of 1.00 g cm⁻³, this would mean the mass of the solution would be 150 g.

Therefore, the thermal energy can be calculated as

$$Q = mc\Delta T$$

$$Q = (150)(4.18)(43)$$

$$Q = 26\,961 \text{ J}$$

To find the change in enthalpy we will need to calculate the number of moles. The substance indicated in the question is copper(II) sulfate (because zinc is in excess) so we can determine the number of moles of CuSO₄ using the concentration and volume ensuring the volume is used in dm³.

$$n = CV$$

$$n = (1.0) \left(\frac{150}{1000} \right)$$

$$n = 0.150 \text{ mol}$$

Now we have all information needed to calculate the change in enthalpy of the reaction:

$$\Delta H = -\frac{Q}{n}$$

$$\Delta H = -\frac{26\,961}{0.150}$$

$$\Delta H = -179\,740 \text{ J mol}^{-1}$$

Finally, we convert to kJ mol⁻¹ and report the result to an appropriate number of significant figures

$$\Delta H = -\frac{179\,740}{1000}$$

$$\Delta H = -180 \text{ kJ mol}^{-1}$$

Accounting for heat loss during data processing

When you consider using a calorimeter to determine the change in enthalpy for a reaction, what do you think the largest source of error would be? The system might be closed but it is not isolated; therefore, it is always 'connected' to the earth in some way, making heat loss unavoidable. This heat loss results in smaller changes in temperature than we would expect theoretically. This means that experimentally, we would record a final temperature lower than theoretically possible for exothermic reactions and a final temperature higher than theoretically possible for endothermic reactions.

Can you think of any ways we can reduce heat loss experimentally? Insulation of calorimeters is one way to reduce heat loss. This can be done by surrounding the calorimeter with insulating materials, increasing the thickness of these materials and minimising contact with the air.

Can you think of any ways we might be able to process data to account for this heat loss? Let's look at how the temperature changes during the reaction of excess solid zinc reacting with a 150 cm³ solution of 1.0 mol dm⁻³ copper(II) sulfate in **Figure 2**.

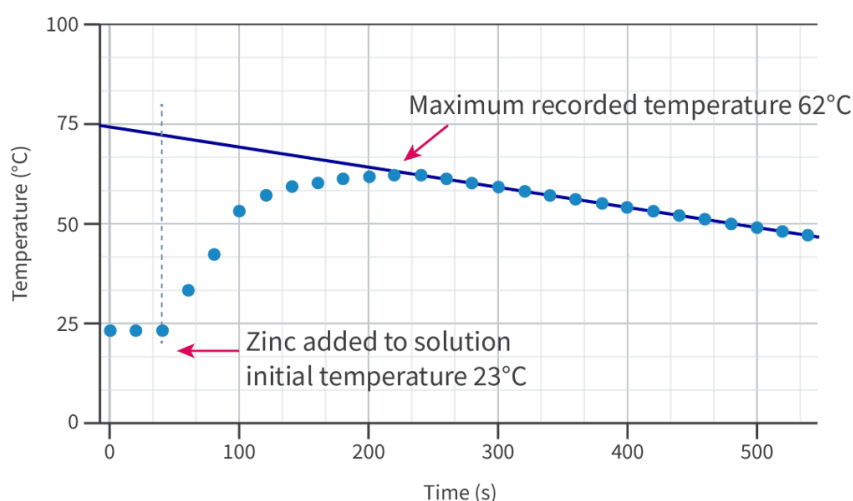


Figure 2. Temperature changes during the reaction of solid zinc with aqueous copper(II) sulfate.

© More information for figure 2

In **Figure 2**, we can see how it takes time before we are able to record the maximum temperature of this exothermic reaction. During this time, heat would be constantly lost to the air, calorimeter and unreacted zinc. Ideally, heat would be lost to these factors at the same rate. This would mean that heat would be lost linearly while the solution cools. We can take advantage of the linear rate of cooling to estimate the theoretical maximum temperature. **Figure 3** shows how this might be done.

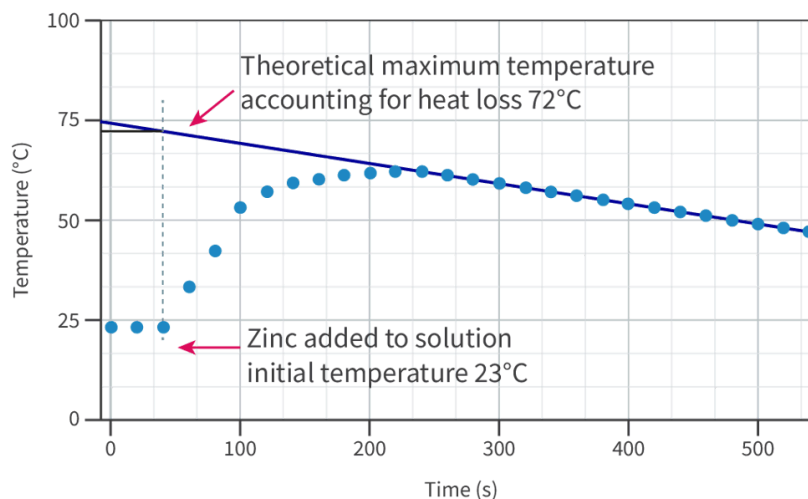


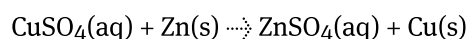
Figure 3. Extrapolation to account for heat loss to determine a theoretical final temperature.

📄 More information for figure 3

Figure 3 shows how by extrapolating the line representing the rate of cooling back to the point where the reaction started can provide a good estimate of the theoretical final temperature. We now have all information needed to solve for this change in enthalpy of reaction. Try it yourself first, and then check with the solutions in **Worked example 4**.

Worked example 4

Calculate the enthalpy change of reaction for the reaction of



Given that excess solid zinc reacted with 150 cm³ of 1.0 mol dm⁻³ copper(II) sulfate and showed the temperature changes shown in **Figures 2 and 3**.

To calculate the enthalpy of reaction we need to make use of the following two equations.

$$\Delta H = -\frac{Q}{n} \text{ and } Q = mc\Delta T$$

The extrapolated value for the maximum theoretical temperature would need to be used when calculating the temperature change:

$$\Delta T = 72 - 23$$

$$\Delta T = 49^\circ\text{C}$$

The aqueous solution would experience the measured temperature rise of 49°C, therefore the mass and specific heat capacity of the aqueous solution can be assumed to be that of water.

As the volume of the aqueous solution is 150 cm³ and water has a density of 1.00 g cm⁻³, this would mean the mass of the solution would be 150 g.

Therefore, the thermal energy can be calculated as

$$Q = mc\Delta T$$

$$Q = (150)(4.18)(49)$$

$$Q = 30723 \text{ J}$$

To find the change in enthalpy, we will need to calculate the number of moles. The substance indicated in the question is copper(II) sulfate (because zinc is in excess) so we can determine the number of moles of CuSO₄ using the concentration and volume ensuring the volume is used in dm³.

$$n = CV$$

$$n = (1.0) \left(\frac{150}{1000} \right)$$

$$n = 0.150 \text{ mol}$$

Now we have all information needed to calculate the change in enthalpy of the reaction:

$$\Delta H = -\frac{Q}{n}$$

$$\Delta H = -\frac{30723}{0.150}$$

$$\Delta H = -204\,820 \text{ J mol}^{-1}$$

Finally, convert to kJ mol⁻¹ and report to an appropriate number of significant figures:

$$\Delta H = -\frac{204\,820}{1000}$$

$$\Delta H = -205 \text{ kJ mol}^{-1}$$

Aspect: Measurement

Producing knowledge in thermochemistry is possible because of assumptions made. These assumptions also limit the validity of the knowledge produced. There are benefits and limitations associated with making assumptions and these must be evaluated and justified for any scientific investigation.

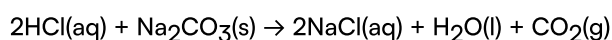
The following includes a summary of assumptions made in determining the change in enthalpy of reaction for zinc and copper(II) sulfate:

- All energy released from the reaction is transferred to the water solvent.
- The solution is dilute therefore negligible volume from solute.
- The excess zinc powder and produced copper solid do not absorb any of the heat released from the reaction.
- Solution loses heat to the surroundings linearly.
- Water has a density of 1.00 g cm^{-3} .

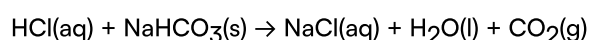
Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication skills
 - Presenting data appropriately
 - Using symbols and communication conventions consistently
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

When hydrochloric acid, HCl, reacts with sodium carbonate an exothermic reaction occurs.



When hydrochloric acid, HCl, reacts with sodium hydrogencarbonate an endothermic reaction occurs.



The temperature data can be monitored throughout each of these reactions using a calorimeter. The following table shows an example of experimental temperature data collected from a student's experiments. Each experiment involves the reaction of excess carbonate with 75.0 cm^3 of 1.50 mol dm^{-3} HCl(aq).

Time (s)	Temperature (°C) for reaction of sodium carbonate with hydrochloric acid	h
0	22.0	
20	22.0	
40	22.0	
60	22.0	
80	25.0	
100	28.0	
120	32.0	
140	35.0	
160	37.0	
180	38.0	
200	39.0	
220	40.0	
240	39.0	
260	39.0	
280	39.0	
300	38.0	
320	37.0	
340	36.0	
360	35.0	

Time (s)	Temperature (°C) for reaction of sodium carbonate with hydrochloric acid	h
380	34.0	
400	33.0	
420	32.0	

For each reaction, complete the following:

- Plot temperature data and carry out an extrapolation to determine the theoretical temperature accounting for heat loss.
- Calculate the change in enthalpy.
- Sketch an energy profile diagram.
- Describe what the system and surroundings are in these experiments.